



Physics 1

Grade 11 · Course #2003380 · Full Year (Semesters 1 & 2) · Optima Academy Online

Family Course Overview

A one-page guide to the year ahead

Physics 1 introduces scholars to the principles that govern **motion, forces, and energy** — and to the discovery that nature can be described, analyzed, and predicted with mathematics. Students solve real problems weekly, visualize motion and forces in VR, run simulations, and build from kinematics all the way to circuits and electric fields. **Prerequisites: Chemistry and Algebra 1.**

THE YEAR AT A GLANCE

Q1 Motion (Kinematics)

~8 weeks

Describing motion in one and two dimensions: position, velocity, and acceleration; free fall and projectile motion; then launching into 2D motion — with unit quizzes and a test along the way.

Q2 Forces & Newton's Laws

~9 weeks

Angled launches, then the heart of mechanics: identifying and drawing forces, mass and acceleration, friction, inclined planes, apparent weight, tension systems, and hanging masses.

Practice-rich: 2–4 problem-solving assignments weekly, from "Will it Move?" to multi-force systems at angles.

Q3 Energy, Momentum, Gravitation & Heat

~9 weeks

Work and energy, kinetic and potential energy with friction, momentum and collisions, universal gravitation (Kepler's laws), and specific heat — with weekly class discussions.

Project: Rube Goldberg Machine — "Energy in Action" — plus simulation labs like the PhET Energy Skate Park ramp challenge.

Q4 Waves, Optics & Electricity

~10 weeks

Waves, sound, and the Doppler effect; optics with mirrors, lenses, and ray diagrams; then circuits, power and charge, static electricity, and electric fields — ending with the final exam.

Hands-on: students build and solve series, parallel, and mixed circuits, including VR circuit problems.

A TYPICAL WEEK

- ▶ **Read:** Quick Reference Guides that introduce each concept and worked examples
- ▶ **Discuss:** a weekly class discussion connecting physics to the real world (esp. Semester 2)
- ▶ **Solve:** 2–4 problem-solving assignments, labs, or simulations
- ▶ **Explore:** VR visualizations of motion, forces, and circuits on Optima's virtual campus
- ▶ **Check in:** quizzes and tests at the end of each unit

TIME COMMITMENT

250 minutes / week ≈ 4 hrs 10 min

- ▶ Lessons & reference guides: ~50 min
- ▶ Problem sets, labs & simulations: ~135 min
- ▶ VR activities: ~35 min
- ▶ Discussions, quizzes, review: ~30 min

Weekly minutes come from Optima's master schedule; the per-component split is an estimate. Project and test weeks may run longer.

WHAT STUDENTS NEED

- ▶ Scientific calculator (graphing recommended)
- ▶ Physics notebook for notes, labs & problem-solving
- ▶ VR headset for Optima's virtual campus
- ▶ Art supplies for projects & models (lists provided in advance)
- ▶ **On AI:** Optima embraces AI as a tool that enhances — never replaces — learning, guided by the seven Pillars of Virtue

ON-DEMAND (ASYNCHRONOUS)

This course is designed for self-paced learning: students work through the weekly modules on their own schedule with a teacher available for support, feedback, and grading.

LIVE SCHOOL

Live-school students complete the same course in VR every school day, together with a teacher and classmates in real time.

PLEASE NOTE

This overview describes the course as designed. The week-to-week experience may vary from teacher to teacher, as each teacher leans into their own strengths and passions.

Optima Academy Online · Grading: 60% summative (tests, quizzes, projects/labs) · 40% formative (daily assignments & participation)